

Zihao Chen

mikechen012345@uchicago.edu | (773) 851-1204 | mikechen012345.github.io | Chicago, IL

Education

The University of Chicago	Chicago, IL
<i>B.S. in Computer Science; B.S. in Neuroscience (GPA: 3.98/4.0)</i>	06/2026
Courses: AI Agent, Machine Learning, Software Development, System Programming, Computer Security	

Research Experience

Research Assistant (Advisor: Prof. Nicholas Hatsopoulos)	Chicago, IL
<i>Hatsopoulos Lab - University of Chicago</i>	06/2024 – present
- Engineered a fault-tolerant, modular, and extensible Python pipeline to stream, synchronize, and store real-time Internet of Thing (IoT) data at high throughput (~80 GB/session at 200 MBps). - Developed a computer vision and data processing workflow using OpenCV and PyTorch to perform feature extraction and noise reduction on large-scale datasets. Implemented data filtering and Principal Component Analysis (PCA) to reduce feature dimensions from 256 to 25 while improving the model prediction accuracy to 80%. - Trained and evaluated multiple machine learning models (LDA, SVM, Kalman Filter) for time-series prediction to enhance predictive performance and robustness. - Presented research findings at multiple symposia and a major international conference (Society for Neuroscience).	
Research Assistant (Advisor: Prof. Shannon Heald)	Chicago, IL
<i>Attention, Perception, & EXperience (APEX) Lab - University of Chicago</i>	09/2023 – 05/2024
- Optimized a Neuro Evolution of Augmenting Topologies (NEAT) program by incorporating Compositional Pattern-Producing Networks (CPPN), significantly boosting performance and the complexity of the output. - Developed automated data validation and cleaning scripts using Pandas for large Electroencephalogram (EEG) datasets, reducing manual data correction effort by 80% and improving downstream model performance.	

Teaching Experience

Teaching Assistant	Chicago, IL
<i>CMSC 25300/35300 Mathematical Foundation of Machine Learning</i>	09/2025 – present
- Led discussion sections for 23 students, covering topics in linear algebra, probability, and machine learning algorithms. - Designed and implemented weekly code demos and assignments for 150+ master's and undergraduate students, directly impacting the computational and theoretical skill development of the cohort. - Provided one-on-one mentorship during weekly office hours, effectively debugging student code and clarifying theoretical concepts for successful assignment completion and deep learning comprehension.	
Teaching Assistant	Chicago, IL
<i>CMSC 14100/14200 Introduction to Computer Science</i>	01/2024 – 06/2025
- Evaluated Python assignments for 20-25 students weekly, providing targeted feedback on software development. - Collaborated with professors to refine grading rubrics and course materials based on student performance trends.	

Presentations

Chen, Z., Sobinov, A., Raman, C., & Hatsopoulos, N., (2025, November 18). <i>Encoding of grasp force in human motor cortex during object transport</i> [Paper presentation]. Society for Neuroscience 2025, San Diego, CA, United States. - Asclipe, M., Codol, O., Sobinov, A., Chen, Z., Park, J., Hatsopoulos, N., Dudman, J., Gallego, J., Lajoie, G., & Perich, M., (2025, October 16). <i>Evolutionarily preserved neural dynamics across species in the motor cortex during motor behaviors</i> [Manuscript submitted for conference]. Computational and Systems Neuroscience 2026, Lisbon, Portugal. Chen, Z., Sobinov, A., Raman, C., & Hatsopoulos, N., (2025, April 25). <i>Exploring missing grasp force signals in motor cortex during object transport</i> [Poster session]. QUAD Research Symposium, Chicago, IL, United States. Chen, Z., Sobinov, A., Raman, C., & Hatsopoulos, N., (2024, August 13). <i>Decoding grasp force in the M1 during object transportation</i> [Paper presentation]. Jeff Metcalf's Research Fellowship Symposium, Chicago, IL, United States.	
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Selected Research Projects

AI-Powered Resume Creation and Optimization Application

07/2024 – 12/2024

- Architected an AI-powered application to automate the resume lifecycle, from data ingestion and multi-version LaTeX rendering to AI-driven resume analysis and content generation.
- Engineered a custom **Large Language Model** (LLM) framework to implement **tool calling**, structured output, and self-consistency, enabling LLMs to perform live web scraping, data verification, and local file system interaction.
- Benchmarked Qwen-2.5-32B against Llama-3.1-70B and GPT-3.5, selecting it for its superior native support for structured output and efficient performance on consumer 24GB GPUs.
- Leveraged the GPT-4 API with prompt engineering to increase high-impact action verbs by 57%.

AI-Powered Full-stack Personal Issues Consultant

10/2025 – present

- Architected and developed a full-stack, multi-agent AI consultation platform designed to guide users through complex personal decision-making (e.g., career planning, relationship management, strategic recommendations).
- Engineered multi-agent workflow leveraging **LangChain** and **LangGraph** with per-agent memory and external tool-calling capabilities, enabling agents to dynamically store and retrieve context as well as perform online searches for real-time relevant resource suggestions and grounding.
- Designed and delivered the entire application stack using a **Django** REST API backend and a responsive **React** frontend.
- Integrated a data pipeline using **PostgreSQL** for user authentication and long-term agent memory persistence, and **Milvus** as a high-performance vector database to power efficient embedding storage and retrieval.

Honors & Awards

- Enrico Fermi Scholars in the Physical Sciences Collegiate Division (Top 5% in the past 5 years), 2025, The University of Chicago
- Janet Rowley Scholars in the Biological Sciences Collegiate Division (Top 5% in the past 5 years), 2025, The University of Chicago
- Phi Beta Kappa Academic Honor Society, 2025, The University of Chicago
- University of Chicago QUAD Research Scholarship, 2025, The University of Chicago
- Jeff Metcalf's Neuroscience Research Fellowship, 2024, The University of Chicago

Leadership Experience

Software Development Team Leader

Chicago, IL

ChiGame Gaming Platform Development Team

10/2023 – 12/2023

- Led a team of 23 students in a 93-person class to Develop a full-stack gaming platform for the UChicago community, featuring user authentication, real-time direct messaging, and a dynamic tournament management system with automated, schedule-based phase progression.
- Guided the development of a backend service within an **Agile** framework, covering the full software development life cycle from requirement analysis and system design to final deployment on **Google Cloud Platform** (GCP).
- Facilitated inter-group communication for development progress synchronization with in-person weekly meetings.
- Managed **Git** version control for the 23-person team, reviewing 50+ pull requests to ensure code quality.

Software Development Team Leader

Chicago, IL

UC OpenData Club

03/2023 – 06/2023

- Led a team of 5 students to architect and launch a full-stack web application to provide real-time operating hours for 30+ university buildings, libraries, and dining halls, serving over 100 unique student users within the first month.
- Developed a **Python**-based web scraping pipeline with Beautiful Soup and scheduled it to automate daily data collection from disparate university sources, ensuring 99.5% data accuracy and eliminating manual updates.
- Designed a responsive frontend with **Django**, reducing user search time by 60% compared to navigating official university websites, and deployed the application on **AWS** for scalable, serverless hosting.

Skills

Programming Languages: Python, C, C++, Java, SQL, R, MATLAB

Technologies: PyTorch, Langchain (ReAct Agent), MCP, Vector Databases (Milvus, Qdrant), Pandas, NumPy, Scikit-learn, AWS, GCP, Django, FastAPI, Docker, Git, Linux